Raw Materials, Compositions, Equipment, and Manufacturing Processes for Vehicle Brake Pad Manufacture

1. Introduction

Vehicle brake pads are composite friction materials engineered to convert kinetic energy into heat through friction while maintaining stable performance across a wide range of temperatures, pressures, and environmental conditions. The manufacture of brake pads involves the precise selection of raw materials, careful formulation of compositions, and a multi-stage production process that transforms powdered ingredients into a dense, structurally sound friction element bonded to a steel backing plate.

Modern brake pads are classified into three main categories: **Non-Asbestos Organic (NAO)**, **Low Steel**, and **Semi-Metallic** formulations . Each class employs distinct raw material strategies, but all share the same fundamental component categories: binders, fibers, fillers, lubricants, and abrasives .

2. Raw Materials and Compositions

2.1 Binders

The binder is the matrix that holds all other components together and provides structural integrity to the friction material. The type and quantity of binder critically determine the fade resistance and thermal stability of the brake pad .

Phenolic resins are the most widely used binders in brake pad manufacture. They may be used in solid powder form or as liquid resoles . Phenolic resins offer excellent heat resistance, dimensional stability, and the ability to wet and bond various fillers and reinforcements.

Rubber-modified phenolic resins, including nitrile rubber, natural rubber, and styrene-butadiene rubber, are also employed to improve flexibility and impact resistance . The binder typically constitutes **16–24% by volume** of the friction material .

2.2 Fibers and Reinforcements

Fibers provide mechanical strength, thermal stability, and rigidity to the friction composite. They carry a greater load than other constituents and help resist cracking under thermal stress .

Common fiber types include:

- **Aramid fibers** (e.g., Kevlar): High temperature resistance, used as web-forming fibers
- **Mineral fibers**: Rock wool, wollastonite, glass fiber
- **Metallic fibers**: Copper, brass, steel, bronze
- **Carbon fibers**: For high-performance applications
- **Cellulose and polyacrylonitrile pulps**: Web-forming organic fibers

In NAO formulations, fibers typically constitute **4–12% by volume**, while the total fiber content (including web-forming fibers) may range from **22–60% by volume** in specialized non-asbestos pads .

2.3 Fillers

Fillers serve to improve manufacturability, reduce cost, and modify specific properties such as density, porosity, and thermal conductivity .

Inorganic fillers include:

- **Barium sulfate (barite)**: The most commonly used filler, with a high melting point (1580°C) and excellent chemical stability

- **Calcium carbonate, magnesium oxide, mica, vermiculite**

- **Petroleum coke, carbon black**

Organic fillers include:

- **Cashew dust**: Provides viscoelastic properties, reduces noise and vibration

- **Rubber powder**: Nitrile rubber or tire tread rubber powder, improves damping characteristics

Fillers typically constitute **46–65% by volume** of the friction material .

2.4 Lubricants and Friction Modifiers

Lubricants stabilize the friction coefficient, particularly at high temperatures, and protect the counterface from excessive wear while reducing noise and vibration .

Common lubricants include:

- **Graphite**: Naturally occurring solid lubricant

- **Metal sulfides**: Antimony trisulfide (Sb_2S_3), molybdenum disulfide (MoS_2), tin sulfide, copper sulfide, zinc sulfide

- **Metal oxides**: Magnetite (Fe_2O_3), zinc oxide (ZnO), chromium oxide (Cr_2O_3)

Lubricants typically constitute **2–5% by volume** of the friction material .

2.5 Abrasives

Abrasives increase the friction coefficient and aid in removing undesirable surface coatings from the brake disc during braking, thereby improving braking effectiveness .

Common abrasives include:

- **Zirconium silicate, zirconium oxide**

- **Alumina (Al_2O_3), silicon carbide (SiC)**

- **Mica, quartz, magnesia (MgO)**

Abrasives are selected based on hardness, fracture toughness, shape, size, and aggressiveness against the counterface material . They typically constitute **10–22% by volume** .

2.6 Representative Formulation

A representative NAO brake pad formulation (by volume) is shown below :

Component	Material	Volume %
Binder	Phenolic resin	21
Fiber	Aramid	7
Lubricants	Antimony trisulfide + metal sulfide blend	4
Abrasives	Mineral fiber + MgO + mica	15
Fillers	Rubber powder, graphite, petroleum coke, barytes, friction dust, hydrated lime	53

3. Equipment

3.1 Mixing Equipment

High-shear mixers such as Henschel, Loedige, or Eirich type blenders are used to achieve homogeneous mixing of the diverse raw materials . These mixers must provide sufficient shear to break up agglomerates while avoiding excessive heat generation that could prematurely initiate curing.

Roll blenders may be employed for hot blending, where raw components are passed between heated counter-rotating rollers to soften or liquefy the binder without initiating polymerization . This produces a semifinished product in the form of chips, ribbon, or sheet.

3.2 Forming Equipment

Hydraulic molding presses are central to brake pad manufacture. Typical pressing conditions include:

- **Pressure**: 2,000 PSI (approximately 14 MPa) or higher
- **Temperature**: 285°F (140°C) for preforming, or 265–295°F (130–146°C) for final pressing
- **Pressure for final pressing**: 5–50 tons per piece

For non-asbestos pads produced via aqueous slurry routes, molds with water-permeable walls are used to compress the slurry and remove water, forming a damp preform .

3.3 Curing and Post-Curing Equipment

Curing ovens are used for the final cure of the binder. Typical conditions include:

- **Temperature**: 330–370°F (165–188°C) for baking
- **Post-curing**: 150–400°C for 10 minutes to 10 hours

Infrared heating systems with near-infrared heaters (peak wavelength 1200–1700 nm) may be used for sintering, as these wavelengths penetrate deeply into the stack of friction materials for uniform heating .

3.4 Finishing Equipment

Grinding machines are used to achieve final thickness tolerances and surface finish. Robotic arms with grippers and sanding/grinding wheels are increasingly employed for automated finishing operations . **Scorching** may be performed using laser systems to burn off surface black materials and create a porous scorched layer for improved initial friction performance .

3.5 Automated Production Equipment

Modern brake pad manufacturing increasingly employs automated systems. Automatic production equipment may include material feeding assemblies, hot pressing molds, finished product handling plates with electromagnets, cleaning devices with brushes, and steel backing feeding mechanisms .

4. Manufacturing Processes

4.1 Mixing and Blending

The first step involves mixing the raw materials to achieve a homogeneous blend. A typical sequence involves:

1. Premixing the binder with fibers to ensure uniform dispersion
2. Adding fillers, lubricants, and abrasives
3. Final mixing under controlled conditions to avoid premature curing

The mixing process may be conducted at room temperature (cold blending) or at elevated temperature (hot blending) depending on the binder system. Hot blending with roll blenders produces a semifinished product that can be stored and subsequently shaped .

4.2 Preforming

The mixed friction material is formed into a preform of the desired shape. This may be accomplished by:

- **Cold pressing**: Pressing the powder at room temperature under 4–25 tons/piece pressure

- **Aqueous slurry preforming**: For non-asbestos pads, an aqueous slurry containing binder, fibers, and fillers is fed into a mold with water-permeable walls, compressed to remove water, and dried to form a damp preform

The preform has sufficient green strength for handling but is not fully cured.

4.3 Bonding to Backing Plate

The friction preform is bonded to a steel backing plate, which provides structural support and mounting provisions. The backing plate is typically:

- Degreased and primer-treated

- Placed in the mold with the friction material

- Bonded through the simultaneous application of heat and pressure

The bonding process may involve co-pressing the support plate with the friction mixture, followed by curing .

4.4 Hot Pressing and Curing

The assembly is subjected to hot pressing to consolidate the friction material to near-final density and to initiate curing of the binder. Typical conditions:

- **Pressure**: 5–50 tons/piece

- **Temperature**: 265–295°F (130–146°C)

The hot pressing step consolidates the preform to substantially the finished density without completing the cure .

4.5 Post-Curing (Baking)

The consolidated pad is baked to fully cure the binder. Typical conditions:

- **Temperature**: 330–370°F (165–188°C)

- **Post-cure**: 150–400°C for 10 minutes to 10 hours

For specialized systems, pyrolysis may be conducted at temperatures below 800°C, preferably 400–600°C, in an inert atmosphere (argon or nitrogen) with a heating rate of 4–6°C/min and a hold time of 3–5 hours .

4.6 Finishing Operations

After curing, the brake pad undergoes finishing operations:

- **Grinding**: To achieve final thickness and parallelism

- **Chamfering**: To reduce noise and improve initial bedding

- **Scorching**: Laser or thermal treatment to burn off surface organics and create a porous surface layer

The scorching step typically involves laser energy density of 150–2000 kW/m² and processing time of 0.5–60 seconds .

4.7 Quality Control and Painting

Finished brake pads are inspected for dimensional accuracy, friction performance, and structural integrity. They may be spray-painted or powder-painted and kiln-dried before packaging .

5. Summary

The manufacture of vehicle brake pads is a multi-stage process requiring careful selection of raw materials, precise formulation, and controlled processing conditions. The key elements are:

Stage	Materials	Equipment	Conditions
Mixing	Binder, fibers, fillers, lubricants, abrasives	Henschel/Loedige blender	Room temp or hot blending
Preforming	Mixed friction material	Hydraulic press	4–25 tons/piece
Bonding	Friction preform + steel plate	Mold press	Heat + pressure
Hot Pressing	Assembly	Hydraulic press	130–146°C, 5–50 tons
Curing/Baking	Consolidated pad	Curing oven	165–188°C
Finishing	Cured pad	Grinding machine, laser	Dimensional control

The trend in modern brake pad manufacturing is toward greater automation, with robotic handling, automated mold cleaning, and integrated quality control systems improving consistency and productivity .